

gdTAI V4.2 T/NK-restricted reintegration

Generated 2026-08-17. This is a development-sidecar analysis, not a classifier result or model promotion.

Refined T/NK pool: 3,927,924 of 4,023,462 cells retained (97.63%); 95,538 cells removed. scVI used 4,000 features and GPU only.

Updated conclusion: no NK training core is accepted. Harmonized productive-chain metadata identify TRA or TRB in 373,339/475,953 boundary cells (78.44%) and in 11,526/20,375 primary NK annotation anchors (56.57%). The previous clusters 0/3/5 candidate-core interpretation is withdrawn.

1. What changed

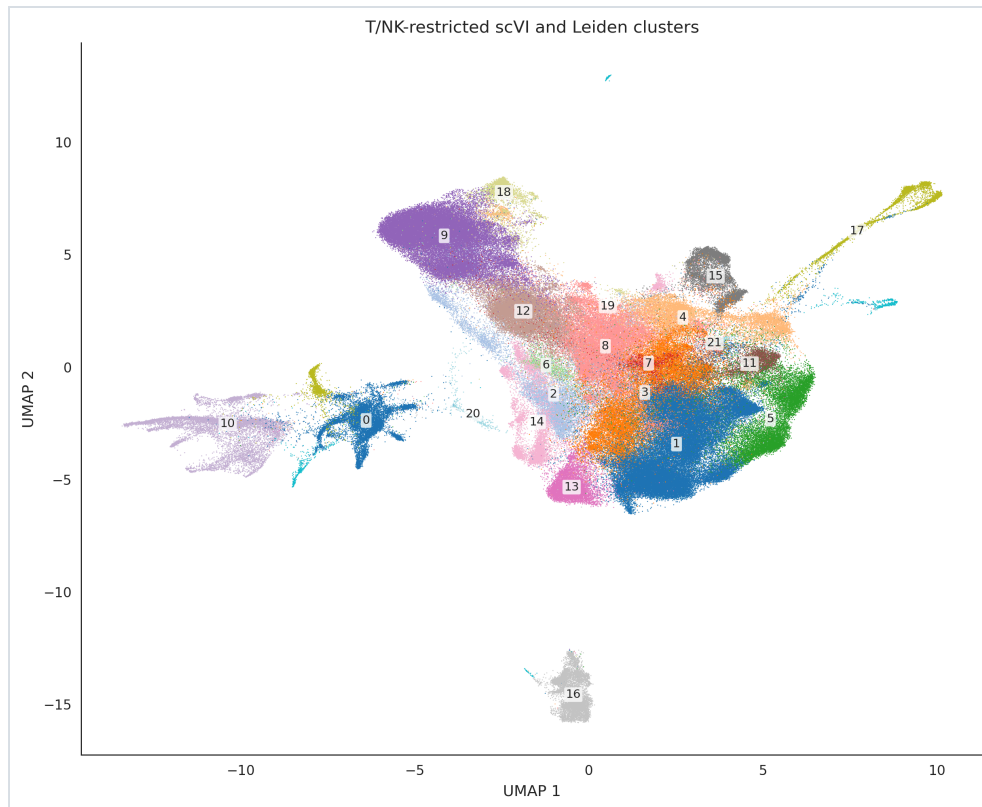
The inputs were already high-recall T/NK candidate objects, but they retained ambiguous off-target cells. This second pass preserves all non-doublet productive-TCR cells and primary NK anchors, then requires direct T-lineage, gamma-delta, NK-receptor, or multigene NK-adaptor-plus-cytotoxic evidence. Cells with broad myeloid evidence and no direct T/gdT/NK-receptor support are removed.

TRDC+ / delta-V- / weak-CD3 is not used as an NK label. It is retained as a phenotype for unsupervised review. Dropout, ambient RNA, and true NK biology remain competing explanations.

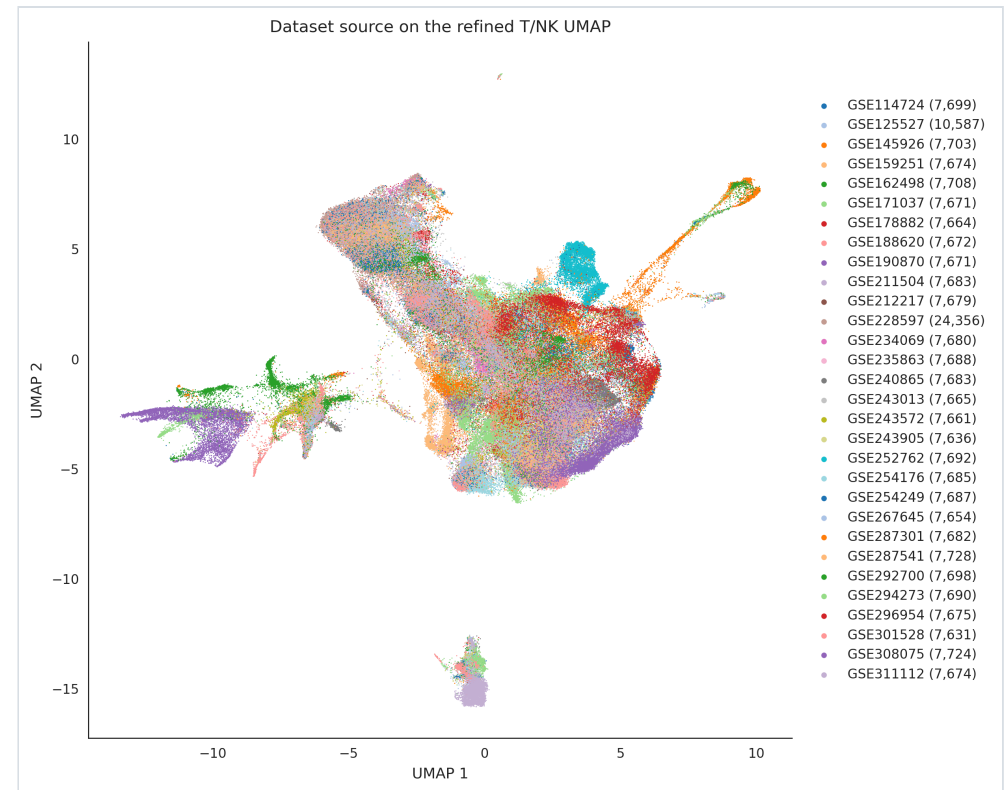
tnk_gate_keep	tnk_gate_reason	n_cells
False	removed_myeloid_conflict	28195
False	removed_no_direct_tnk_evidence	67343
True	gd_lineage	31730
True	nk_receptor	75573
True	primary_nk_protected	21054
True	productive_tcr_protected	967149
True	shared_cytotoxic_plus_nk_adaptor	9004
True	t_lineage	2823414

2. Recomputed HVGs and integration

HVGs were recomputed only after T/NK subsetting with source-balanced sampling. TCR V/J/D genes were excluded from the HVG ranking to limit clonotype-driven structure. A fixed 27-gene CD3/TCR-constant and T/NK lineage panel was forced into the 4,000-feature set. The fit completed in 25.5 minutes on NVIDIA A100 80GB PCIe.

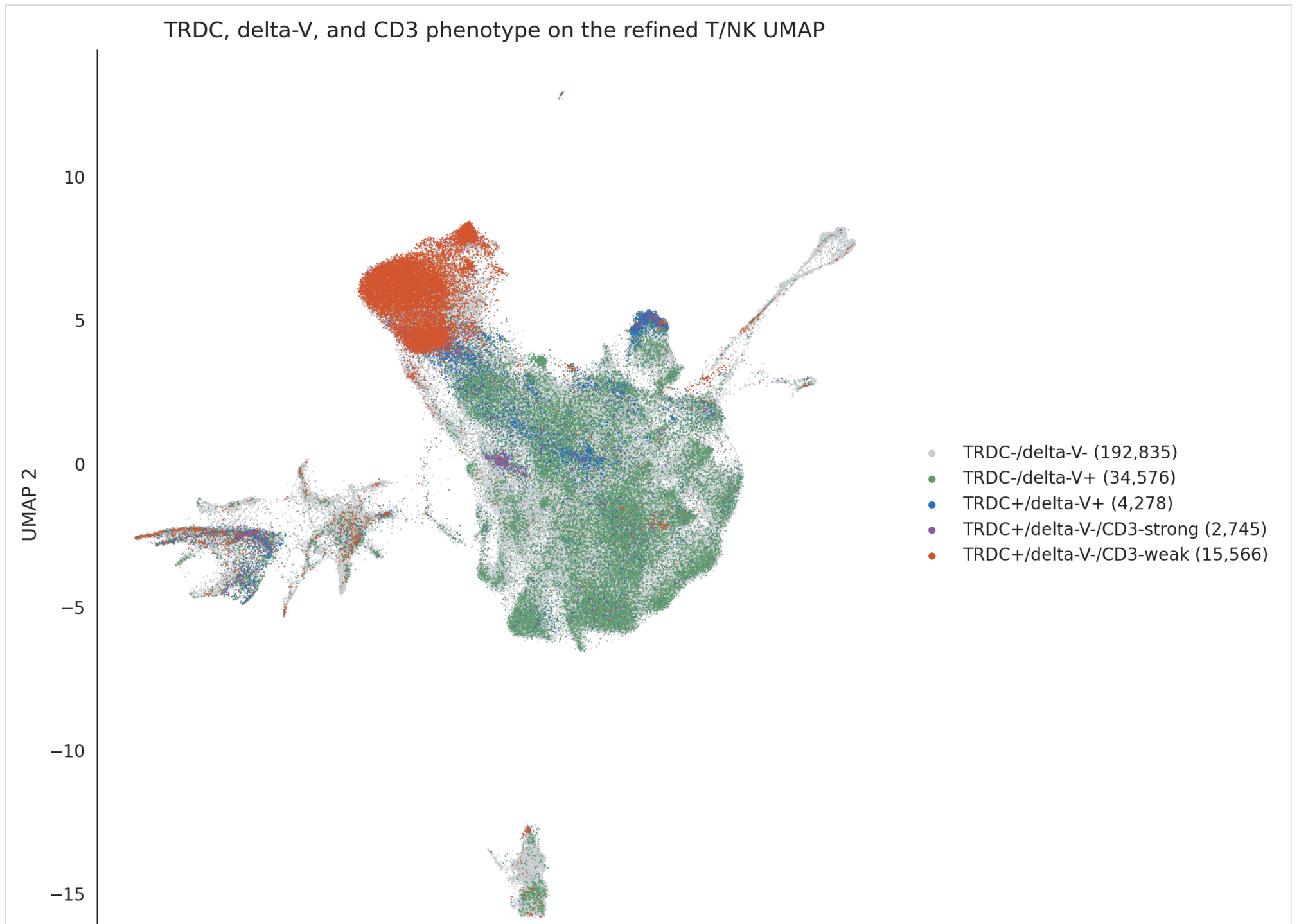


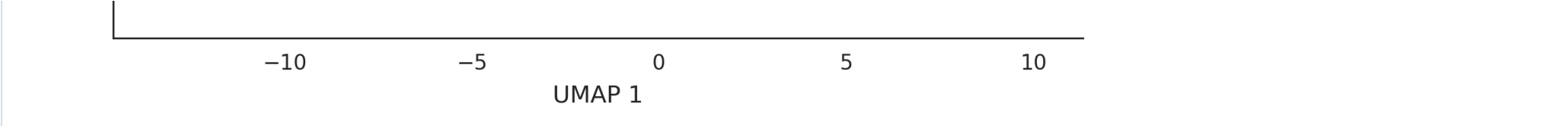
Unsupervised refined clusters.



Dataset mixing remains visible and must be considered in annotation.

3. TRDC, delta-V, and CD3 phenotype



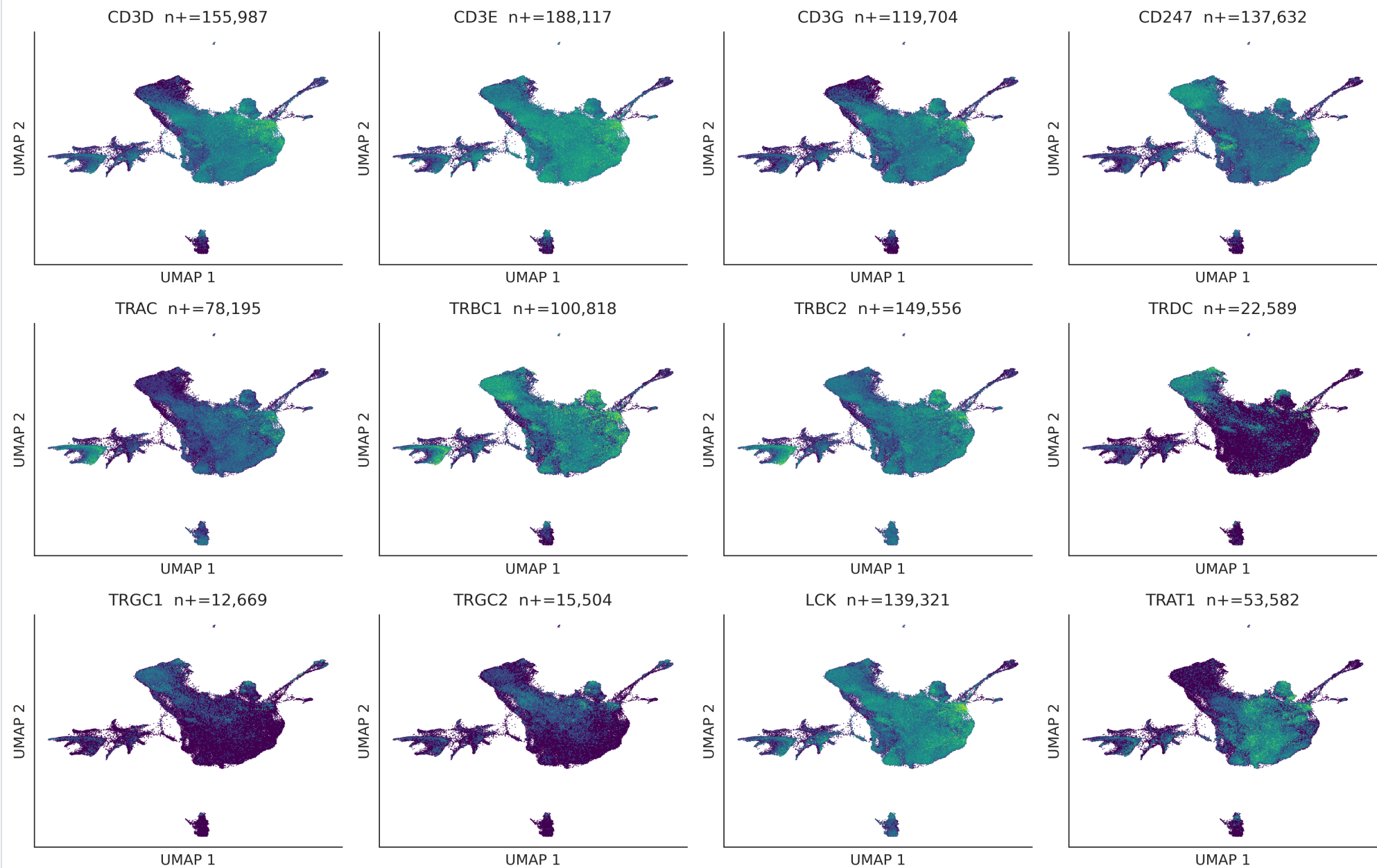


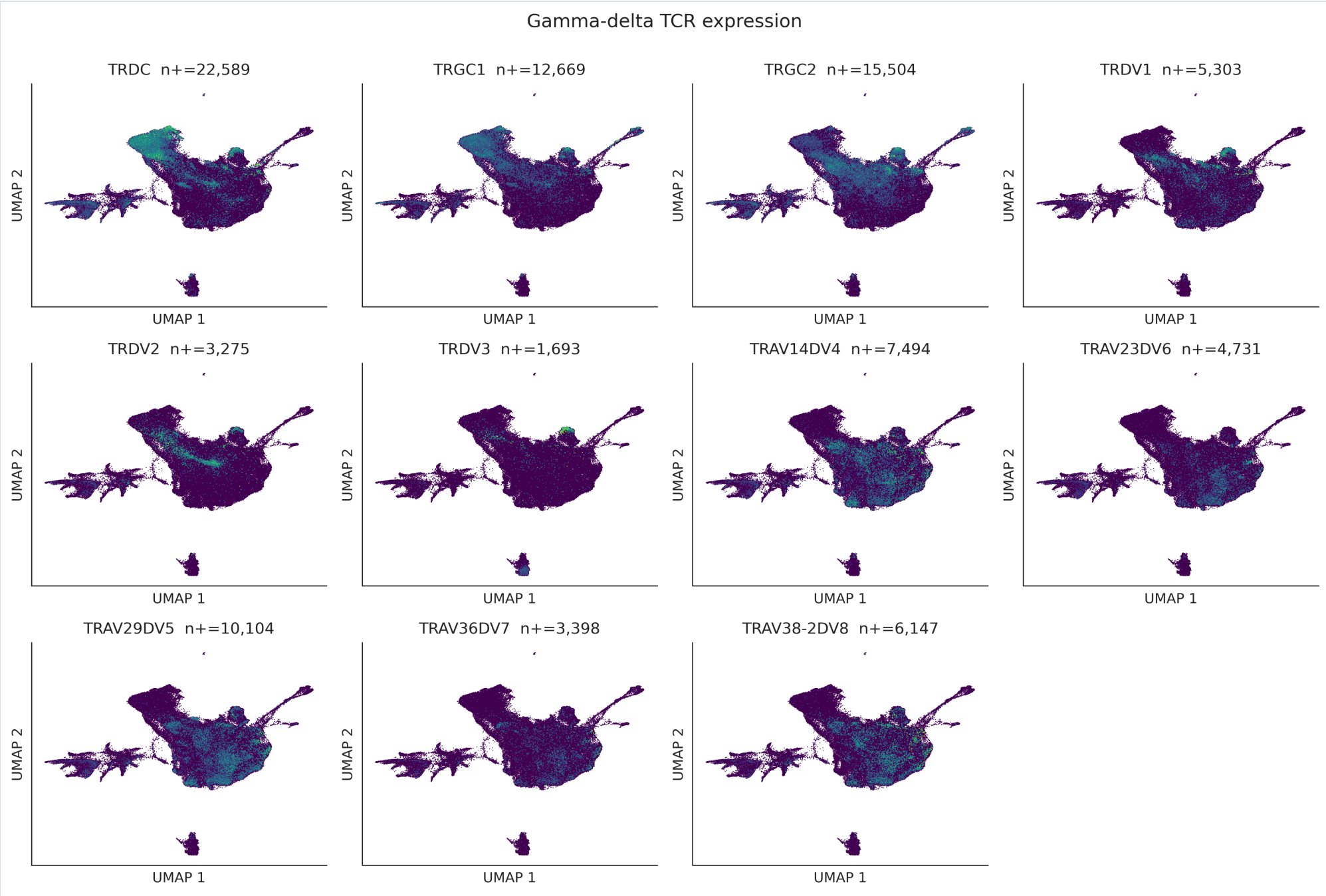
The proposed NK-like phenotype is displayed but not treated as truth.

trdc_delta_v_cd3_phenotype	tnk_gate_keep	n_cells
TRDC+/delta-V+	True	67672
TRDC+/delta-V-/CD3-strong	True	43942
TRDC+/delta-V-/CD3-weak	True	178674
TRDC-/delta-V+	True	585051
TRDC-/delta-V-	False	95538
TRDC-/delta-V-	True	3052585

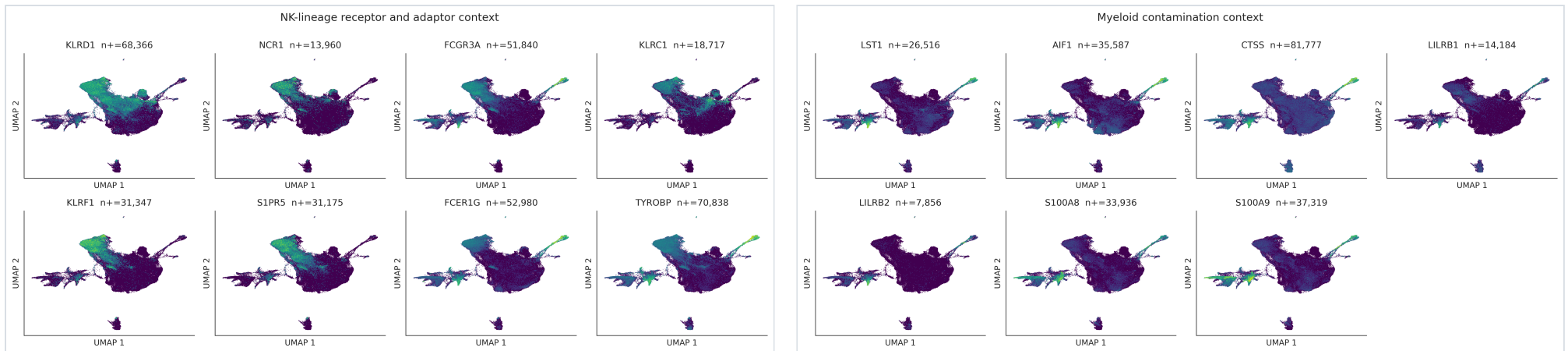
4. CD3 and TCR genes

CD3 and TCR constant-chain expression

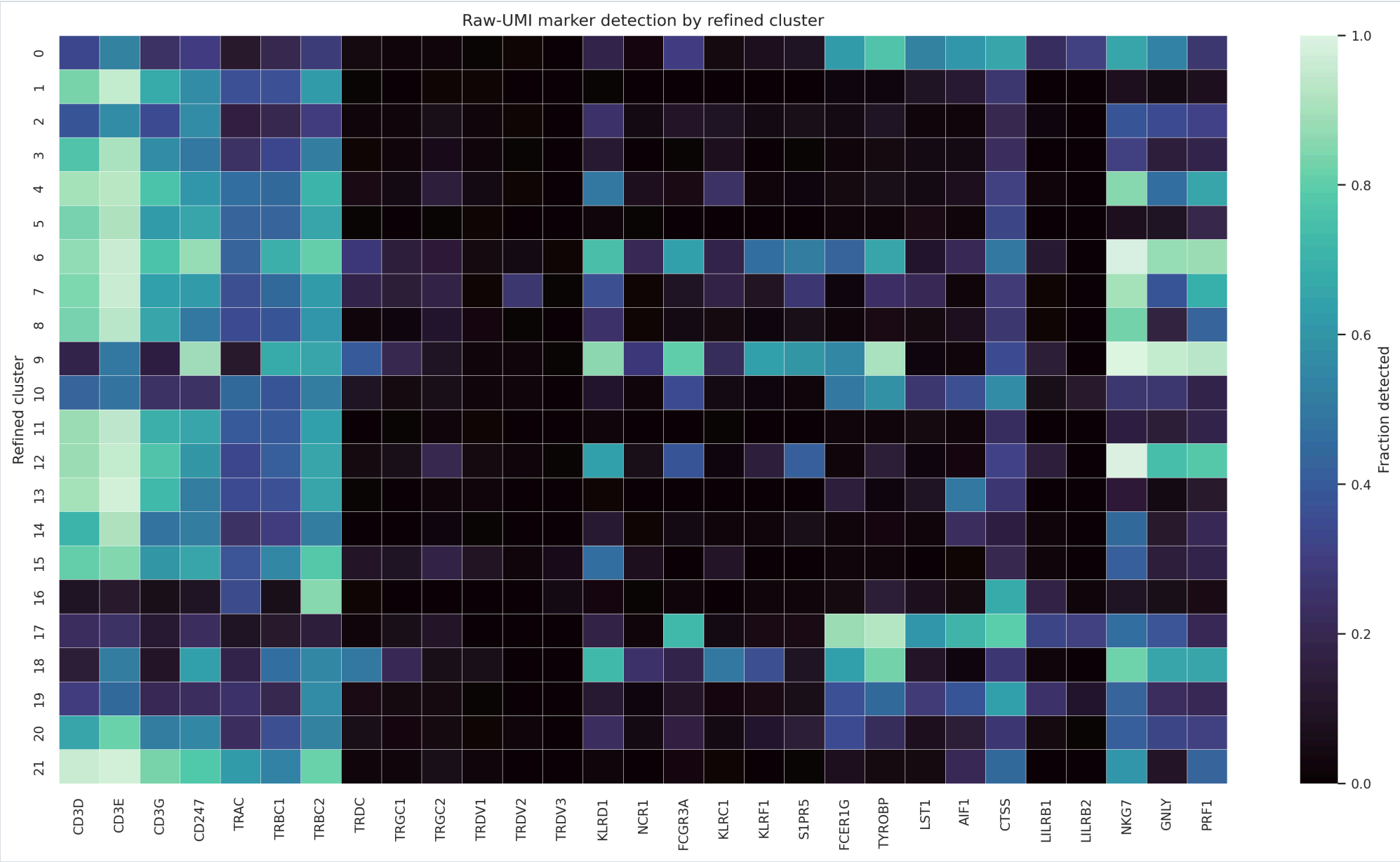




5. NK and contamination context



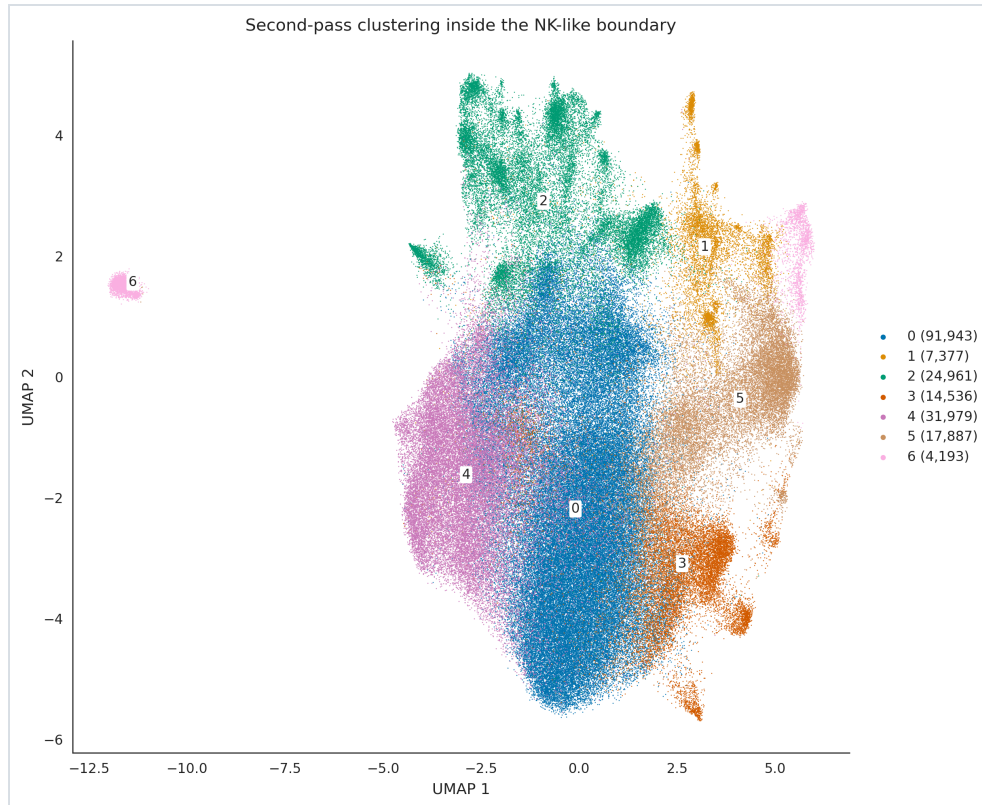
6. Cluster-level evidence



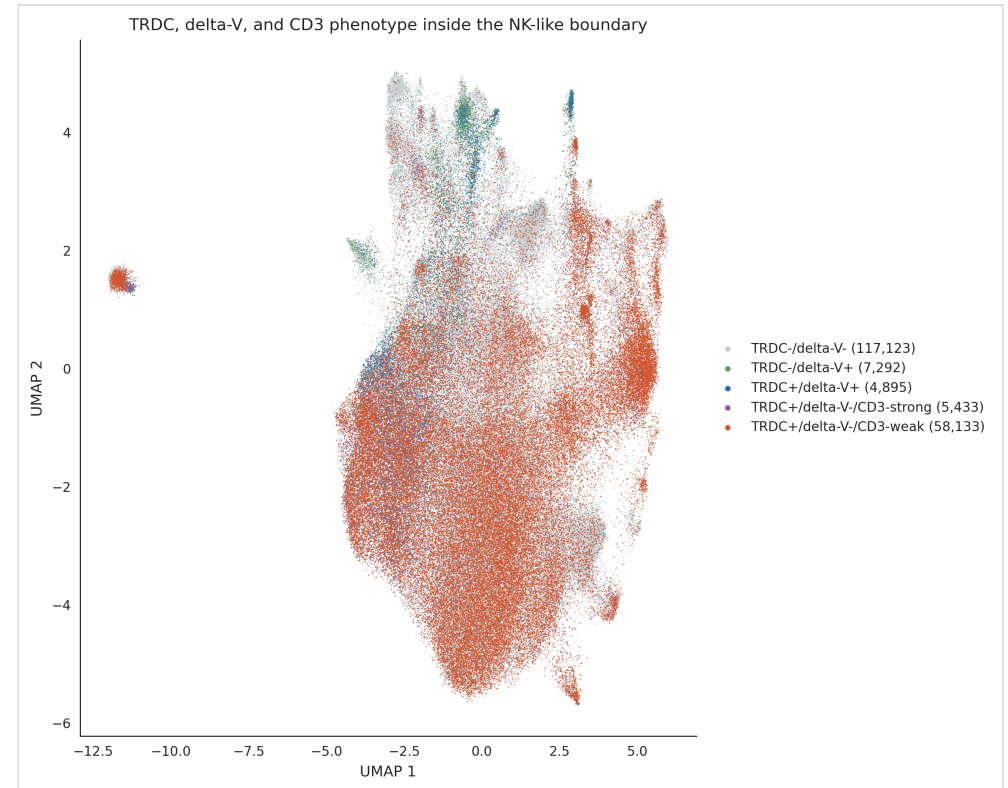
cluster	n_cells	n_primary_nk	n_productive_tcr	primary_nk_fraction	productive_tcr_fraction
9	406026	19227	7894	0.0474	0.0194
18	69927	1148	1354	0.0164	0.0194
2	223236	320	9578	0.0014	0.0429
6	33418	21	14178	0.0006	0.4243
12	295726	174	102309	0.0006	0.3460
4	224352	128	38595	0.0006	0.1720
20	9737	5	4723	0.0005	0.4851
7	69952	13	17636	0.0002	0.2521
16	108584	7	5584	0.0001	0.0514
0	169253	6	28408	0.0000	0.1678
8	419801	5	108993	0.0000	0.2596
5	284518	0	84517	0.0000	0.2971
1	751551	0	365763	0.0000	0.4867
3	432234	0	59380	0.0000	0.1374
13	124524	0	42509	0.0000	0.3414
11	105635	0	18144	0.0000	0.1718
10	37433	0	2997	0.0000	0.0801
14	66044	0	49321	0.0000	0.7468
17	33987	0	1556	0.0000	0.0458
15	45839	0	376	0.0000	0.0082
19	11753	0	1448	0.0000	0.1232
21	4394	0	1886	0.0000	0.4292

7. Second-pass NK-boundary clustering

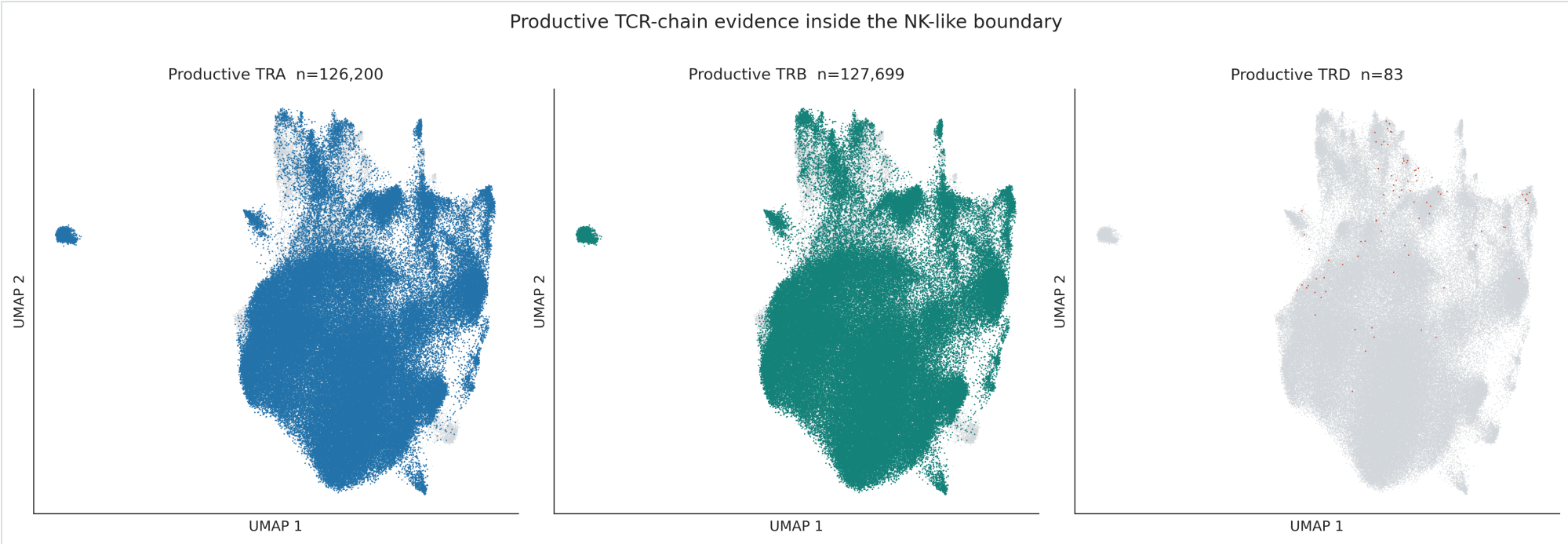
Refined clusters 9 and 18 contain 475,953 cells and recover 20,375/21,054 primary NK annotation anchors (96.77%). The original sidecar mask counted only 9,248 productive-T controls because many chain fields had not been propagated into `TNK_cleaned.h5ad`. The harmonized metadata audit now identifies 373,339 cells with productive TRA or TRB (78.44%) and 83 with productive TRD. Nine GPU Leiden runs were performed inside this boundary; the frozen review partition `boundary_r0.4_s29` contains 7 subclusters. At resolution 0.4, the mean adjusted Rand index across three seed pairs is 0.962 (minimum 0.952).



Second-pass unsupervised subclusters.



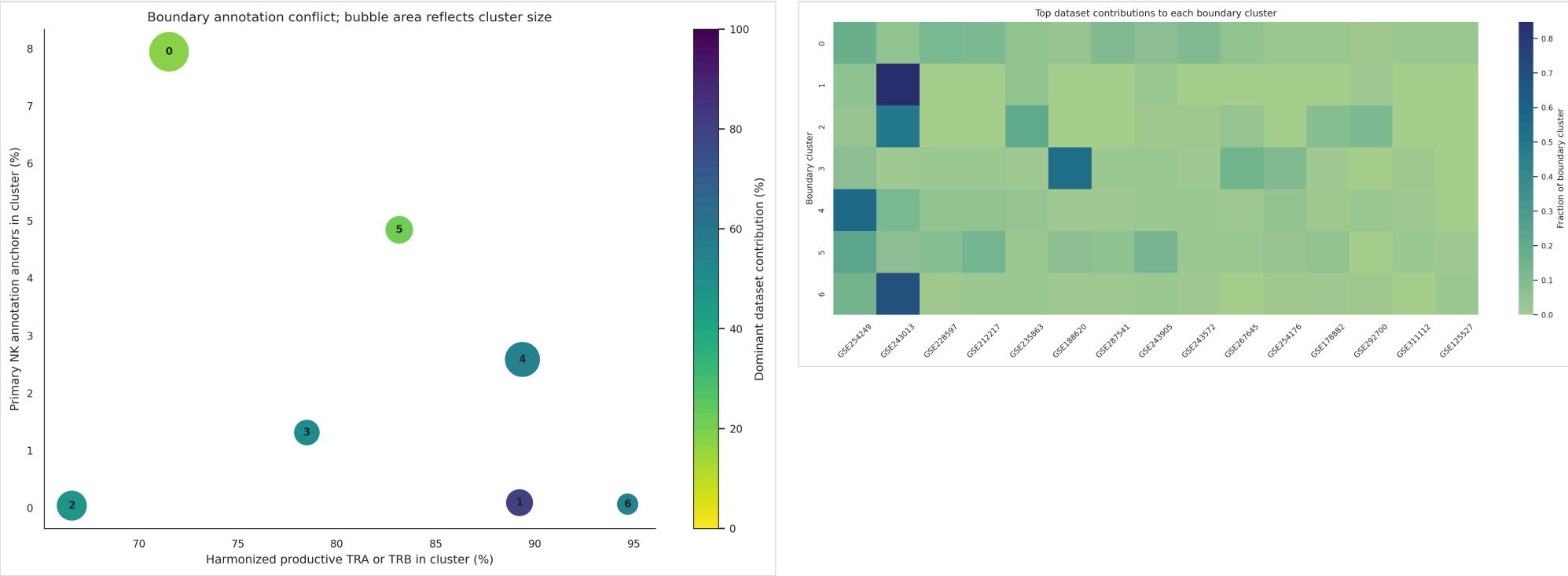
The weak-CD3 phenotype is enriched but is not an NK label.



Chain-specific productive TCR evidence is defined by a nonempty harmonized productive-filtered CDR3, not by RNA expression. Absence of a plotted chain is not informative in libraries without the corresponding V(D)J assay.

chain	n_boundary_productive	fraction_boundary_productive	n_umap_sample_productive	fraction_umap_sample_productive
TRA	371616	0.7808	126200	0.6543
TRB	373186	0.7841	127699	0.6621
TRD	83	0.0002	83	0.0004
TRA_or_TRB	373339	0.7844	127832	0.6628

The productive-chain overlay overturns the earlier boundary interpretation. Harmonized TRA/TRB calls are a strict superset of the source-H5AD calls: all 6,307 source-H5AD-positive boundary cells agree, while 364,091 additional current-atlas cells have productive TRA/TRB metadata that had not been propagated into the H5AD. In addition, 11,526/20,375 (56.57%) primary NK annotation anchors carry productive TRA or TRB, so those annotations cannot independently validate NK identity.



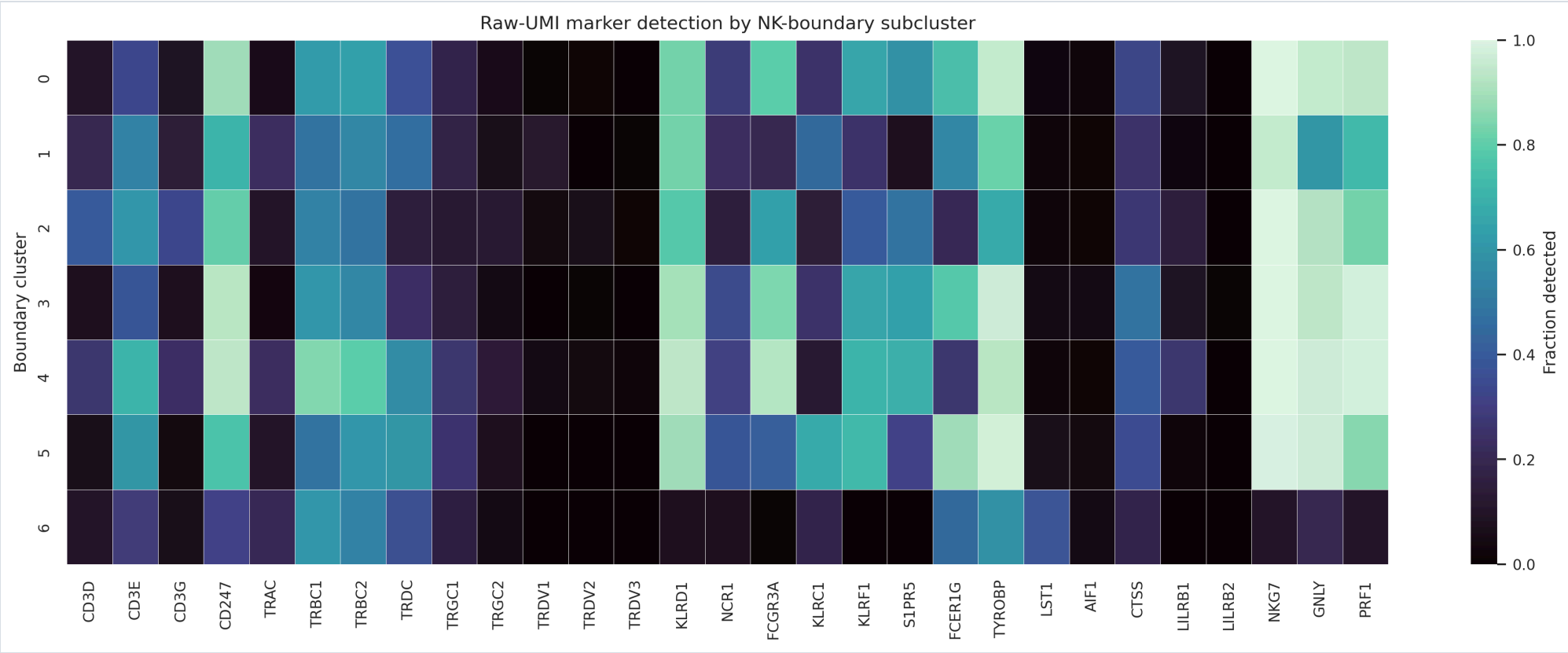
resolution	n_seed_pairs	mean_adjusted_rand_index	min_adjusted_rand_index	max_adjusted_rand_index
0.2000	3	0.9399	0.9115	0.9959
0.4000	3	0.9617	0.9521	0.9671
0.8000	3	0.8450	0.7852	0.9055

8. Revised NK-boundary interpretation

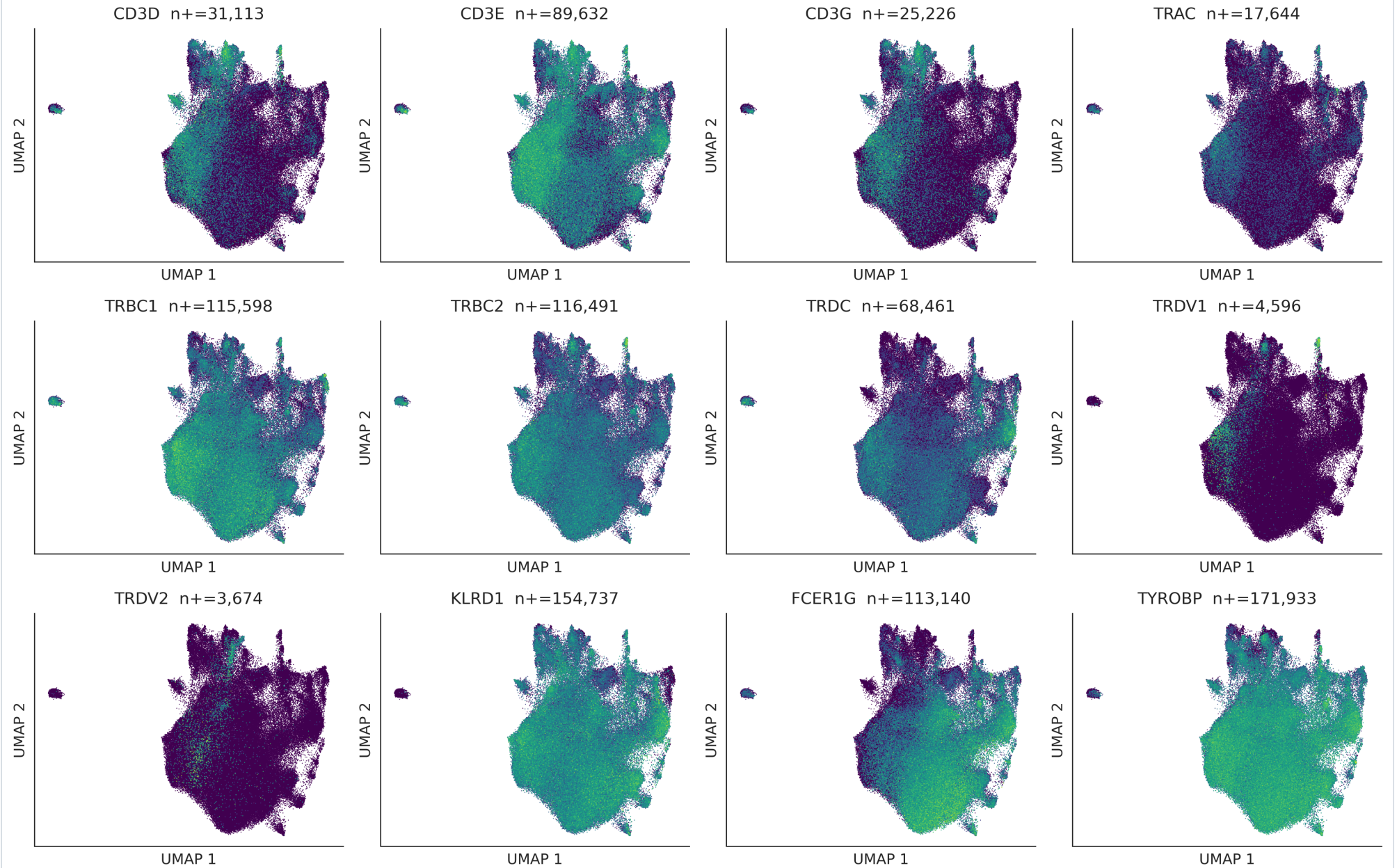
Boundary clusters 0, 3, 5 were initially considered a transcriptomic NK-like core because they combine weak `CD3D/CD3G` , high NK receptors, `FCER1G/TYR0BP` , and cytotoxic genes with low `LST1/AIF1` . The productive-chain audit rejects that interpretation: 190,757/257,569 cells (74.06%) have productive TRA or TRB, while only 19 have productive TRD. The previous candidate-core designation is therefore withdrawn; these clusters are NK-like transcriptomically but TCR-alpha-beta dominated.

No boundary subcluster is promoted as NK training truth. The transcriptomic similarity of cytotoxic alpha-beta T, gamma-delta T, and NK cells cannot be resolved by TRDC , cytotoxic genes, FCER1G/TYR0BP , or any single marker. The TCR metadata conflict and the compromised primary NK anchors must be repaired before a new classifier iteration.

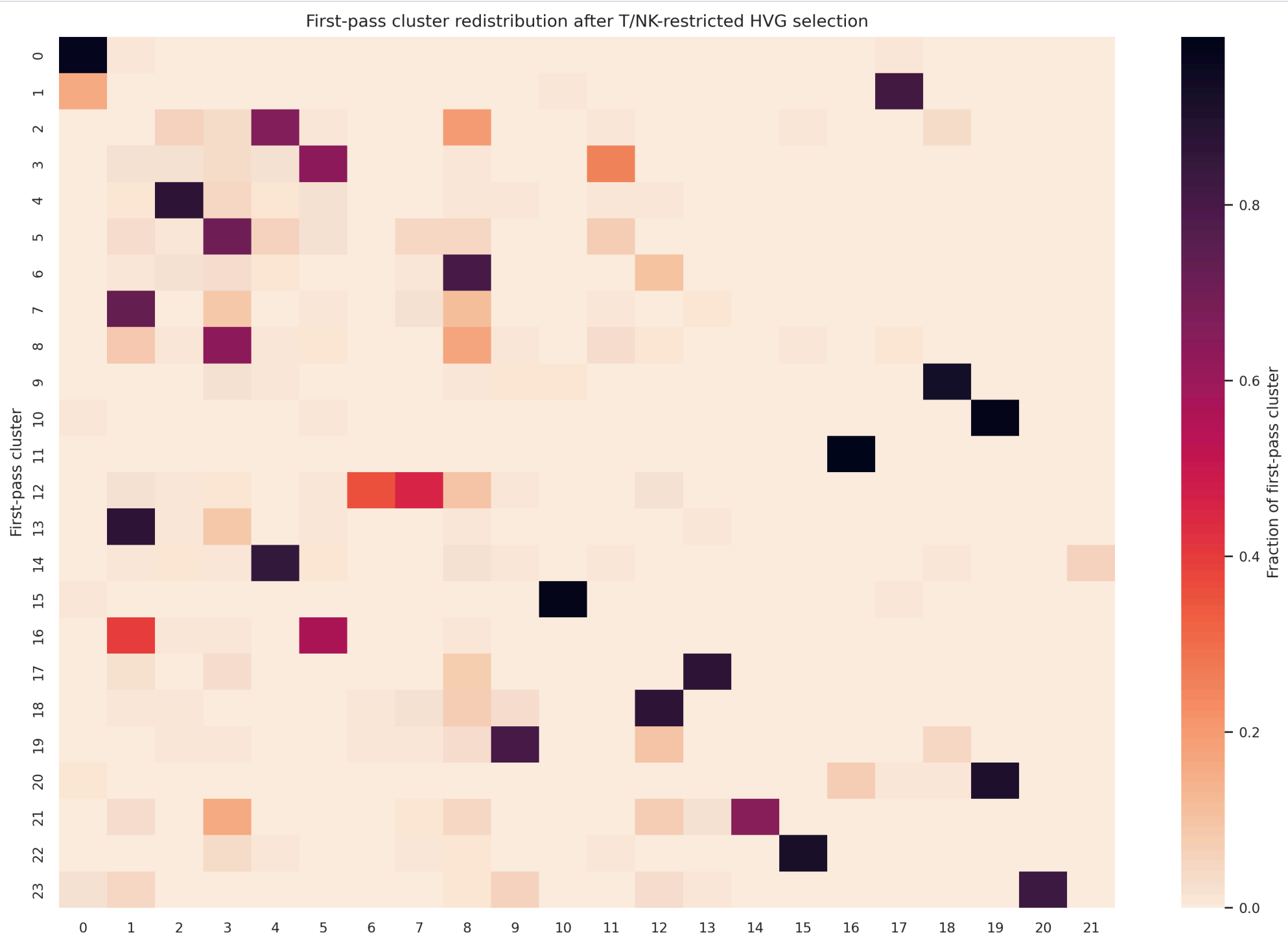
boundary_cluster	review_annotation	n_cells	n_primary_nk	n_sidecar_productive_tcr	n_productive_ab	n_productive_trd	dominant_source	primary_nk_fraction	fraction_productive_ab	dominant_sou
0	NK_like_transcriptome_TCRab_dominated	190265	15112	2274	136092	16	GSE254249	0.0794	0.7153	
1	source_dominated_NK_like_review	35596	31	291	31762	3	GSE243013	0.0009	0.8923	
2	cytotoxic_T_like	57804	20	2563	38507	44	GSE243013	0.0003	0.6662	
3	NK_like_transcriptome_TCRab_dominated	27887	365	626	21888	0	GSE188620	0.0131	0.7849	
4	mixed_T_NK	113540	2932	2200	101477	11	GSE254249	0.0258	0.8938	
5	NK_like_transcriptome_TCRab_dominated	39417	1908	859	32777	3	GSE254249	0.0484	0.8315	
6	off_target_or_low_quality	11444	7	435	10836	6	GSE243013	0.0006	0.9469	



T, gamma-delta, and NK evidence inside the NK-like boundary



9. Difference from the first pass



Refined cluster

Rows are first-pass clusters and columns are refined clusters; values are row fractions.

10. Retained cells by source

source_gse_id	tnk_gate_keep	n_cells	fraction_of_source
GSE243013	True	748188	0.9852
GSE254249	True	397778	0.9786
GSE243905	True	347177	0.9907
GSE287301	True	321965	0.9463
GSE235863	True	237023	0.9804
GSE212217	True	211187	0.9882
GSE243572	True	166787	0.8861
GSE228597	True	149198	0.9959
GSE188620	True	119093	0.9665
GSE308075	True	111333	0.9911
GSE287541	True	108777	0.9811
GSE254176	True	95796	0.9881
GSE267645	True	90323	0.9765
GSE311112	True	89992	0.9933
GSE296954	True	86360	0.9971
GSE292700	True	76208	0.9739
GSE162498	True	73904	0.9740
GSE159251	True	68792	0.9985
GSE178882	True	64856	0.9572
GSE294273	True	55403	0.9939
GSE125527	True	45529	0.9813
GSE252762	True	45003	0.9971
GSE301528	True	41996	0.9522
GSE211504	True	36684	0.9850
GSE171037	True	31721	0.9932
GSE114724	True	28651	1.0000
GSE234069	True	20639	0.9919
GSE240865	True	20563	0.9485
GSE145926	True	20292	0.9645
GSE190870	True	16706	0.9972

11. Interpretation boundary

The integration identifies coherent NK-like and T-like transcriptomic neighborhoods, but the harmonized chain audit shows that the selected boundary is predominantly productive TCR-alpha-beta. The existing primary NK annotation anchors are also compromised by productive TRA/TRB evidence. No boundary cluster is suitable as NK training truth until TCR metadata are propagated consistently, anchor conflicts are resolved, and expression-independent NK labels are reconstructed.

Reproducibility

- Config SHA-256: `ee93d1dfa5505fcb59d1ef76e3ff00a238d3e26a79db3713b9d0b9bbac9f9080`
- Script SHA-256: `0179a4a21f78115705327b37415d414eee01781c63709a8bbec2a38cdece0440`
- Staged H5AD SHA-256: `de69f8fa73e16dc88630850b73763a1e98ae3dbebcbf6a5e3884be8def740e16`
- Latent SHA-256: `2db605d8c10add04d81902e3fb586767e4485fa073d89fc565502ca75328baa0`
- Partitions SHA-256: `ae40cc6933419efd9796cc56704478a3d9563be4aa1c4849ab02824125d8e2a8`
- Boundary partitions SHA-256: `b4137a3aad812ddc2f87eed095ee4c662f2d81297e331fd93c5261c7ee930b14`
- Source H5AD mutation: **none**; classifier fitting: **none**; GitHub push: **none**.